

Spotlight on Daylighting

Daylighting systems, which use natural lighting to supplement electric lighting, have the potential to cut energy use, reduce peak demand, and create a more desirable indoor environment, yet these systems often fail to live up to their potential (see "New Designs in Active Daylighting," *HE* Jan/Feb '06, p. 40). One reason they fail is that they are sensitive to photosensor placement and performance, but there have been no easy-to-use tools to help designers predict performance and determine optimum sensor positioning.

The Sensor Placement and Orientation Tool (SPOT) is a free software package that helps designers establish correct photosensor placement relative to a proposed daylighting and electric lighting design (see Figure 1). It also analyzes and predicts overall system performance prior to field installation and commissioning.

Features and Benefits

Anyone who can use Microsoft Excel can use SPOT. The user interface is an Excel worksheet, while most of the calculations made, using a sophisticated daylighting software program known as Radiance, are invisible to the user. Based on user input, the tool calculates nighttime illumination levels and the range of daylight contribution throughout the year. Advanced users can also access Radiance to further customize the calculations. SPOT accounts for many

variables, such as room geometry, surface reflectances, solar orientation, electric lighting layout, and window design. The major elements include

Project information. To start a project, the user sets the project name and the performance measurement units to be used.

Options. The user has options to direct the program to create renderings, include a variety of shading devices, set reflectivity and rotation angles of blinds, and specify the types of light shelf to be analyzed.

ing performance under a sampling of conditions (see Figure 2); and (3) photosensor placement recommendations for each luminaire zone.

Analysis. The analysis portion of SPOT allows the user to apply various photosensor placements to the luminaire zones, adjust the photosensor system settings, and run annual performance calculations. A Photosensor Analyzer screen allows users to mix and match the various photosensor scenarios defined previously and analyze how they will perform under a larger set of

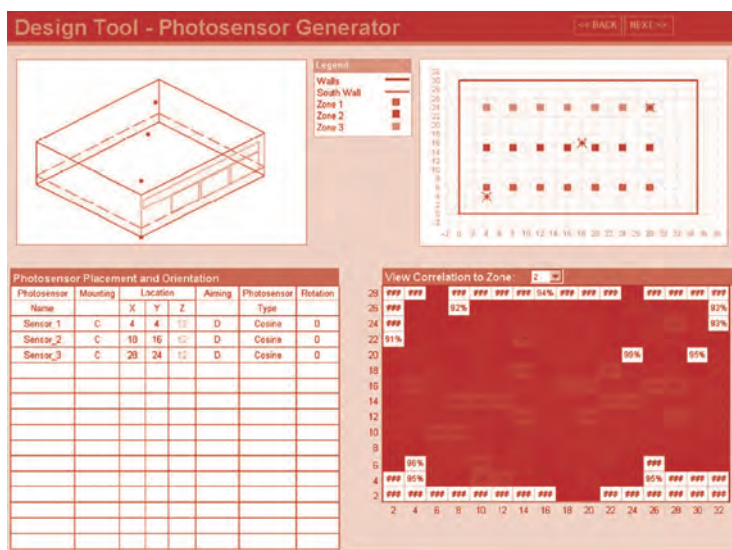


Figure 1. Sensor Placement and Orientation Tool (SPOT) software, a tool for daylighting system analysis and design, can be downloaded from the Internet free of charge. This screen shot shows recommended photosensor locations.

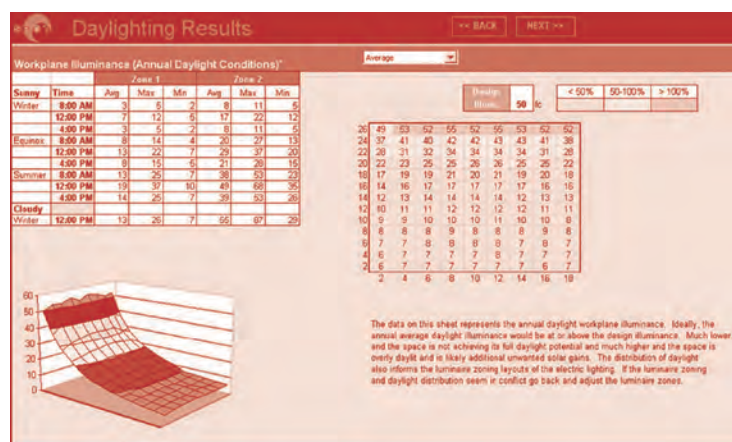


Figure 2. This output screen from SPOT shows the work plane illumination levels as they vary in the course of a day and throughout the year.

Geometry input.

The user defines the room geometry and electric lighting layout for the space to be analyzed. The program lets the user define reflectances of all the surfaces in the room or building and of the ground outside. The user can also specify any overhangs and light shelves to be considered for shading purposes. For electric lighting, the user can define the type of luminaire to be used.

Site and usage.

The user can define the location of the building, set schedules for occupancy, and establish the desired control strategy for the shading device. The program accommodates both weekly and annual schedules.

Design tools.

Once the project has been fully specified, the software moves into the design phase. The design portion of SPOT uses the geometry and site information to provide and report three sets of information back to the user: (1) electric lighting performance under nighttime conditions; (2) annual daylight-

representative days and sky conditions. An Annual Analysis screen provides a more accurate analysis of the photosensor system by accounting for climate conditions, schedules, and time of day.

Applications

SPOT was developed with classrooms in mind, but it can also be applied to other spaces, such as offices or homes. The software handles top and side daylight sources and can model any electric-lighting source from Illuminating Engineering Society (IES) files. The photosensor portion of the program would probably not be very useful for residential applications, but the design tool module allows a designer to analyze how the electric light and daylight will interact in a space throughout the year, for a given glazing configuration and home orientation in a geographic location. The analysis tells the designer whether or not modifications are required to reduce or eliminate glare.

SPOT can be installed on any computer that has

- Windows 2000, NT, or XP;
- Excel 97 (or higher);
- A 400-MHz processor; and
- At least 128 MB of available disk space.

California Codes and Standards

SPOT will help designers to comply more effectively with the daylighting requirements in California's Title 24 energy code for commercial buildings and high rise residential buildings. Title 24 requires separate controls for daylit areas and offers substantial energy budget credits for automatic daylighting controls. The 2005 version of the code favors measures, such as daylighting, that save energy during periods of likely peak demand.

What's Next?

Currently, SPOT allows the user to input only simple, right-angled space geometries. Future versions will allow users to model more complex geometries in other computer-aided design packages, and to import those geometries to the program. Other goals for SPOT include developing an extensive database of the performance of photo-

sensors available from various manufacturers, encouraging widespread use of the free software, setting up demonstration projects, and training users.



—Ira Krepchin

Ira Krepchin is a mechanical engineer and research manager with E Source, a Boulder, Colorado-based energy information service.

This project was conducted for the California Energy Commission's Public Interest Energy Research (PIER) Buildings Program. Architectural Energy Corporation developed the SPOT software.

FOR MORE INFORMATION:

Reports documenting this project and providing more details may be downloaded from the Web at www.archenergy.com/lrp/products/spot.htm.

The software may be downloaded from the Web at www.archenergy.com/SPOT/index.html.

Contacts

Judie Porter
Architectural Energy Corporation
Tel: (303)444-4149
E-mail: jporter@archenergy.com

Michael Seaman
California Energy Commission
Tel: (916)654-4981
E-mail: Mseaman@energy.state.ca.us
Web site: www.energy.ca.gov/pier/buildings